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Machine Learning - V4

Q1: Solution:

The general formula we will use:

$$P(\text{class} | X) = P(\text{class}) \times P(A | \text{class}) \times P(B | \text{class}) \times P(C | \text{class})$$

We need to predict class for new instance.

Step 1: Calculate prior probabilities:

Total positive classes: 5
Total negative classes: 5

$P(\text{positive}) = 5 / 10 = 0.5$
$P(\text{negative}) = 5 / 10 = 0.5$

Step 2: Calculate conditional probabilities:

$P(0 | \text{Positive}) = 2 / 5 = 0.4$
$P(1 | \text{positive}) = 1 / 5 = 0.2$
$P(0 | \text{positive}) = 1 / 5 = 0.2$

$P(0 | \text{negative}) = 3 / 5 = 0.6$
$P(1 | \text{negative}) = 2 / 3 = 0.4$
$P(0 | \text{negative}) = 0 / 5 = 0$

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Step 3: Calculate final scores for each class:

Score(Positive) = P(Positive) x P(A| positive) x P(B| positive) x P(C | positive)

$= 0.5 \times 0.4 \times 0.2 \times 0.2$

$= 0.008$

Score (negative) = P(negative) x P(A |negative) x P(B| negative) x P(C | negative)

$= 0.5 \times 0.6 \times 0.4 \times 0$

$= 0$

Step 4: Final prediction

Positive Score > Negative Score

$\boxed{0.008 > 0}$

Hence, our model predicted that:

A = 0, B = 1, C = 0

Belongs to "**Positive**" class

Q2: Gaussian Probability Density:

To find probabilities; we use Gaussian PDF:

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \times e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n-1}}$$

μ = Arithmetic Mean

Step 1:

$P(\text{male}) = 4/8 = 0.5$

$P(\text{female}) = 4/8) = 0.5$

Step 2: Calculating mean and varience:

Male

Height: $6+5.92+5.58+5.92 \Rightarrow$

$\mu = 5.5855$

$\sigma^2 = 0.0350$

Weight:

$\mu = 176.25$, $\sigma^2 = 122.92$

Foot Size: $\mu = 11.25$, $\sigma^2 = 0.9167$

Female

Height: $\mu = 5.4175$, $\sigma^2 = 0.0972$

Weight: $\mu = 132.5$, $\sigma^2 = 558.33$

Foot-Size: $\mu = 7.5$, $\sigma^2 = 1.67$

Step 3: Calculate Conditional Prob:

For Male:

$$P(\text{Height}_{(6)} \mid \text{Male}) = \frac{1}{\sqrt{2\pi(0.0350)}} \times e^{\frac{(130-176.25)^2}{2\times 122.92}}$$

$$= 1.579$$

$$P(\text{Weight} \mid \text{Male}) = \frac{1}{\sqrt{2\pi \times 122.92}} \times e^{\frac{(130-176.25)^2}{2\times 122.92}}$$

$$= 5.98 \times 10^{-6}$$

$$P(\text{Foot} \mid \text{male}) = \frac{1}{\sqrt{2\pi \times 0.9167}} \times e^{\frac{(8-11.25)^2}{2\times 0.9167}}$$

$$= 1.31 \times 10^{-3}$$

For Female Class:

$$P(\text{Height} \mid \text{Female}) = \frac{1}{\sqrt{2\pi \times 0.0972}} \times e^{-\frac{(6-5.475)^2}{2\times 0.0972}}$$

$$= 0.2235$$

$$P(\text{Weight} \mid \text{Female}) = \frac{1}{\sqrt{2\pi \times 0558.33}} \times e^{-\frac{(130-132.5)^2}{2\times 558.3}}$$

$$= 0.0167$$

$$P(\text{Foot} \mid \text{female}) = \frac{1}{\sqrt{2\pi \times 1.667}} \times e^{-\frac{(8-7.5)^2}{2\times 1.67}}$$

$$= 0.2867$$

Step 4: Calculate final Score:

$$\text{Score (Male)} = 0.5 \times 1.5789 \times 5.98 \times 10^{-6} \times 1.31 \times 10^{-3}$$

$$= 6.18 \times 10^{-9}$$

$$\text{Score (Female)} = 0.5 \times 0.2235 \times 0.0167 \times 0.2867$$

$$= 5.35 \times 10^{-4}$$

$$\boxed{\text{Score(Female)} > \text{Score (Male)}}$$

Finally, Female is Predicted

Q3: Naive Bayes:

Step 1: Calculate probabilities:

$$P(\text{Yes}) = \frac{9}{14} = 0.643$$

$$P(\text{No}) = \frac{5}{14} = 0.357$$

Step 2: Categorical Probabilites:

For Overcast:

$P(\text{overcast} \mid \text{yes}) = 4/9 = 0.444$
$P(\text{overcast} \mid \text{no}) = 0/9 = 0$

For Wind:

$P(\text{False} \mid \text{yes}) = 6/9 = 0.667$
$P(\text{False} \mid \text{no}) = 2/5 = 0.40$

Step 3: Continuous Probalities:

$\cancel{P(\text{False} \mid \text{yes}) = 6/9 = 0.67}$
$\cancel{P(\text{False} \mid \text{no}) = 2/5 = 0.4}$

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$$P(v|c) = \frac{1}{\sqrt{2\pi\sigma^2}} \times e^{\frac{(v-\mu)^2}{2\sigma^2}}$$

Temprature = 60,

Yes values: $\mu = 73$, $\sigma^2 = 38$

$P(\text{Temp}|y) = 0.0070$

No values: $\mu = 79.11$, $\sigma^2 = \cancel{104.36}\ 62.3$

$P(\text{Temp}|\text{no}) = 0.0091$

Humidity: (62)

Yes values: $\mu = 79.11$, $\sigma^2 = 104.36$

$P(\text{Humidity}|\text{yes}) = 0.0096$

No Values: $\mu = 86.2$, $\sigma^2 = 94.7$

$P(\text{Humidity}|\text{no}) = 0.0018$

Step 4: Final Scores:

Score(Yes) = P(Yes) x P(overcast | yes) x P(Temp | yes) x P(Temp | yes) x P(Humidity | yes)

score (Yes) = 0.643 x 0.444 x 0.0096 x 0.667

$= 1.28 \times 10^{-5}$

Score(No) = P(No) x P(overcast | no) x P(temp | no) x P(Humidity | no)

= 0.357 x 0 x 0.0091 x 0.0018 x 0.40

= 0

So, Final Answer = Yes
for → overcasted, temp = 60°, humid.

Yes, He will play tennis